27/07/26 **Aim: Study of various network commands used in Linux & Windows**

Basic Networking Commands :-

(i) arp -a : ARP is short form of address resolution protocol. It will show the IP address of your computer along with the IP address & MAC address of your computer.

(ii) hostname : This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

(iii) ip config /all : This command displays detailed config. information about your TCP/IP connection including Router, Gateway, DNS, DHCP & type of Ethernet adapter in your system.

(iv) nbtstat -a : This command helps solve problems with NetBIOS name resolution (Nbt stands for NetBIOS over TCP/IP).

(v) netstat : (network statistics) netstat displays a variety of statistics about a computer's active TCP/IP connections. It is a command line tool for monitoring network connections both incoming & outgoing as well as viewing routing tables, interface statistics etc. Eg : netstat -r

(vi) nslookup : (name server lookup) is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer, the MX records for a domain, or the NS servers of a domain. nslookup can operate in two modes: interactive & non-interactive
Eg: nslookup www.google.com

(vii) pathping : It is unique to Windows & is basically a combination of the Ping & Tracert commands. Pathping traces the route to the destination address then launches a 25s test of each router along the way, gathering statistics on the rate of data loss along each hop.

(viii) ping : (Packet INternet Groper) command is the best way to test connectivity b/w two nodes. Ping uses ICMP (Internet Control Message Protocol) to communicate to other devices.

1. # ping hostname (ping localhost)
2. # ping ip address (ping 4.2.2.2)
3. # ping fully qualified domain name (ping www.facebook.com)

(ix) Route : route command is used to show/manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface

Some important Linux networking commands

(i) ip : The ip command is one of the basic commands every administrator will need in daily work, from setting up new systems & assigning IPs to troubleshooting existing systems. The ip command can show address info, manipulate routing plus display network various devices, interfaces & tunnels.

ip <OPTIONS> <OBJECT> <COMMAND>

Here are some common use cases for the ip command :-

(a) To show the IP addresses assigned to an interface on your server :-

[root @ server ~] # ip address show

(b) To assign and IP to and interface for e.g., enp0s3:

[root@server ~]# ip address add 192.168.1.254/24 dev enp0s3

(c) To delete an IP on and interface,

ip address del 192.168.1.254/24 dev enp0s3

(d) After the status of the interface by bringing the interface eth0,

(i) online : ip link set eth0 up

(ii) offline : ip link set eth0 down

(e) Alter the status of the interface by enabling promiscuous mode for,

ip link set eth0 promisc on

(f) Add a default route (for all addresses) via the local gateway 192.168.1.254 that can be reached on device eth0;

ip route add default via 192.168.1.254 dev eth0

(g) Add a route to 192.168.1.0/24 via the gateway at 192.168.1.254,

ip route add 192.168.1.0/24 via 192.168.1.254

(h) Add a route to 192.168.1.0/24 that can be reached on device,

eth0 : [root@server ~]# ip route cadd 192.168.1.0/24 dev eth0

(i) Delete the route for 192.168.1.0/24 via the gateway at 192.168.1.254 :-

ip route delete 192.168.1.0/24 via 192.168.1.254

(j) Display the route taken for IP 10.10.1.4 :-

ip route get 10.10.1.4

(ii) ifconfig : The ifconfig command was/is a staple in many sysadmin's tool belt for configuring & troubleshooting networks. It has since been replaced by the ip command discussed above.

(iii) mtr :-

MTR (Matt's Traceroute) is a program with a command-line interface that serves as a network diagnostic & troubleshooting tool. This command combines the functionality of the ping & traceroute commands. Just like a traceroute, the mtr command will show the route from a computer to a specified host. mtr provides a lot of statistics about each hop, such as response time & change percentage. With the mtr comenand, you will get more information about the route & be able to see problematic devices along the way.

If you see a sudden increase in response time, or packet loss, then obviously, there is a bad link somewhere. The syntax of this command is as follows :-

mtr <options> hostname/IP

(a) The basic mtr command shows you the statistics, including each hop (hostnames) with time & loss % :-

mtr google.com

(b) Show numeric IP addresses (if you use -g, you will get IP addresses (numbers) instead of hostnames) :-

mtr -g google.com

(c) Show the numeric IP addresses & hostnames,

mtr -b google.com

(d) Set the no. of pings that you want to send,

mtr -c 10 google.com

(iv) tcpdump :-

The tcpdump command is designed for capturing & displaying packets.

You can install tcpdump with the command below,

duf install -y tcpdump

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Result :-

Hence the experiment to study various network commands used in Linux & Windows is successfully completed.